Sebastian Parakal Jose

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Summary

Software engineer with **3+ years** of experience building backend services, cloud systems and mobile apps that people rely on. Two years at NCR / Global Payments on a 24/7 payments platform for **Mastercard** and **American Express**, working across teams in **Java**, **C++**, **Node.js (NestJS)**, **TypeScript**, **Python**, **Go** and **.NET**. As CTO of MOSAIC, directed the launch of the IFRC World Disasters Report platform to **30,000+** users on day one. Builds on **AWS**, **Azure**, **Kafka** and **Kubernetes**, ships mobile apps in **React Native**, Flutter and Kotlin, and designs multi-agent **Claude Code** workflows with independent verification. Master of IT, Monash (**GPA 4.0/4.0**). **Unlimited work rights in Australia.**

Education

Monash University — Melbourne, Australia

Master of Information Technology – ***GPA 4.0/4.0 (WAM 84.867)*** — *Jul 2024 – Sep 2026*

- High Distinctions in Cloud Computing and Security, Mobile and Distributed Computing Systems, Advanced Database Technology, Internet Applications Development, Software Engineering and Foundations of Data Science

National Institute of Technology Calicut — Kerala, India

Bachelor of Technology in Computer Science and Engineering (First Class) — *2018 – 2022*

Experience

Chief Technology Officer (CTO) — Jan 2026 – Aug 2026

Monash Students for AI with Communities (MOSAIC) — *Melbourne, Australia*

- Directed the production launch of the **IFRC** World Disasters Report platform (wdr26.org), launched globally in Geneva on 5 March and serving **30,000+** users on day one
- Top-three contributor (**110+ commits**) in **Next.js** and **React**: newsletter sign-up via the Brevo SDK and Next.js server actions (with tests), an in-app bug-report flow, Arabic (right-to-left), Russian and French localisation, and launch-day UI
- Contributed to **Ctrl+Alt+Prebunk**, an IFRC game that trains players to spot misinformation, built in **Next.js**, **React** and **TypeScript** with **Zustand** state and Jest/React Testing Library tests: feed interactions, persistence fixes and large-screen layout
- Led development of and built features for **Disinformer**, a cross-platform **Flutter** (Dart) misinformation game for IFRC National Societies on iOS, Android and web, backed by **Firebase**, with offline sync, localisation and analytics
- Rolled out AI-assisted development (**Claude Code**, GitHub Copilot, Cursor), code review and Notion onboarding playbooks, so new joiners shipped in their first week

Technical Lead — Feb 2026 – Jun 2026

Monash University | Industry Experience Studio, ***Hush*** *(hushapp.live)* — *Melbourne, Australia*

- Led a **6-person** team over 3 agile iterations to ship Hush, a sensory-aware navigation app; showcased at the Monash IT Industry Experience Expo (**top 1%** of teams)
- Designed a serverless **AWS** stack (Amplify, **API Gateway**, 11 **Python** Lambdas, EventBridge, ECR, S3, **RDS PostgreSQL**/PostGIS) for **US$22/month**: 6 **REST APIs**, a pgRouting quiet-route engine and an hourly **scikit-learn** crowd forecast
- Ran dev/staging/prod with **GitHub Actions** CI and TDD (**Vitest**, **Playwright**); fixed pentest findings (CORS allow-list, parameterised queries); CloudWatch alarms, backups and handover docs
- Leading the in-progress migration to **Microsoft Azure**: Static Web Apps, **API Management**, **Azure Functions**, Container Apps, Event Grid and PostgreSQL in a private VNet, secured by **Entra ID**

Teaching Associate — Jul 2025 – Present

Monash University — *Melbourne, Australia*

- Mentor **100+** students in Android development (**Kotlin**, Jetpack Compose, MVVM, REST APIs); rated **91%** for satisfaction (SETU)
- Built a multi-agent **Claude Code** marking harness: an orchestrator hands each submission to five sub-agents (code-inspector, report-auditor, rubric-scorer, uniqueness-checker and an independent **verifier**) that flags hallucinations, wrong files and arithmetic errors, with a full audit log

- Built **Python** tooling around the agents for anything countable: PPTX/PDF/DOCX extraction, reference checks against the Semantic Scholar and OpenAlex **APIs**, and a validation gate on marks; workbooks are backed up before every write

Software Engineer Jul 2022 – Jul 2024

NCR Corporation / Global Payments *Hyderabad, India*

- Engineered mission-critical **Java** and **C++** backend systems for a 24/7 payment processor on Linux, handling **Mastercard** and **American Express** transactions across ATM, payment-switch and card-payment flows
- Built services and **REST APIs** across multiple product teams in **Node.js** (**NestJS**), **TypeScript** and **Python**
- Implemented and extended **ISO 8583** messaging pipelines routing high-volume transactions between card networks, acquirers and issuing banks, without disrupting live integrations
- On the **L4 escalation team**, root-caused the hardest production incidents and shipped hotfixes under live constraints, improving system reliability by **20%**
- Built **Splunk** dashboards, queries and alerts for pipeline observability, cutting mean time to detect anomalies by **30%**
- Tuned **Oracle SQL** queries, indexes and stored procedures behind transaction records, cutting peak-load latency by **25%**
- Upgraded and migrated the EMV test automation framework to **.NET 6** (SpecFlow, MSTest), covering contact, contactless and other transaction types across multiple environments on **Azure**, with **Jira** integration, **Docker** on Oracle Linux, and **GitHub Actions** and **Jenkins** pipelines
- Rewrote legacy **Perl** scripts as maintainable **Go** tools, and built .NET utilities for encryption and **DUKPT** key management

Projects

SignalStack | *Python, Apache Kafka, PostgreSQL, Redis, FastAPI, Next.js, Terraform, Prometheus, Grafana*

- Built a self-hosted tech-trend platform ingesting real-time activity from **GitHub** and **Hacker News** through an event-driven **Apache Kafka (KRaft)** pipeline on a single free-tier AWS EC2 instance
- Fault-tolerant consumers with partitioned topics, batched **PostgreSQL** writes and **Redis** deduplication, raising throughput ~**250x** at zero added infrastructure cost
- Served through a 9-endpoint **FastAPI** service with Redis caching and a **Next.js** front end; provisioned with **Terraform**, CI/CD in GitHub Actions, monitored with self-hosted **Prometheus** and **Grafana**

CloudPose | *Python, FastAPI, Docker, Kubernetes, YOLOv8, Locust*

- Profiled and optimised the inference pipeline (image resizing, explicit garbage collection, concurrency control), **cutting memory 35%** and **doubling throughput** across horizontally scaled pods

FridgeFolks | *React Native, iOS App Store*

- Built and published **FridgeFolks**, a **React Native** app on the **iOS App Store** that cuts household food waste: barcode scanning, expiry-date tracking, shared pantries and reminder notifications

Technical Skills

Languages: Python, Java, Kotlin, TypeScript, JavaScript, C++, C#, Go, Dart, SQL, Bash, R
Backend: Spring Boot, Node.js (Express, NestJS), .NET, FastAPI, Flask, REST API design (OpenAPI), Apache Kafka, ISO 8583, microservices
Cloud & Infrastructure: AWS (Lambda, API Gateway, S3, DynamoDB, RDS, EC2, ECR, Cognito, EventBridge, SNS, Amplify, CloudWatch), Azure (Functions, API Management, Static Web Apps, Entra ID), Cloudflare Workers and D1, Docker, Kubernetes, Terraform
Data: PostgreSQL (PostGIS, pgRouting), Oracle SQL, DynamoDB, Redis, Firebase/Firestore, Room, star-schema data warehousing
Frontend & Mobile: React, Next.js, Vue 3, Zustand, Tailwind CSS, React Native, Flutter, Android (Jetpack Compose, MVVM, WorkManager, ML Kit)
Quality & Delivery: GitHub Actions, Jenkins, CodeQL, Dependabot, SonarQube; Jest, React Testing Library, Vitest, Playwright, pytest, MSTest, SpecFlow, Locust
Observability & Security: Splunk, Prometheus, Grafana, CloudWatch; least-privilege IAM, secrets management, CORS and security headers, DUKPT
AI Engineering: Claude Code, sub-agents and multi-agent orchestration, harness engineering, GitHub Copilot, Cursor, YOLO, ONNX Runtime, scikit-learn